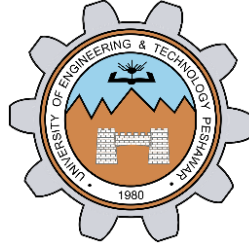


Introduction to ModelSim and Gate Level Modelling

LAB # 01



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CSE-308L Digital System Design Lab

Submitted by: **Ali Asghar**

Registration No.: **21PWCSE2059**

Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Dr. Asif Ali Khan

Date:

24th February 2024

Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

Task 1:

Implement a buffer at the gate level.

Code:

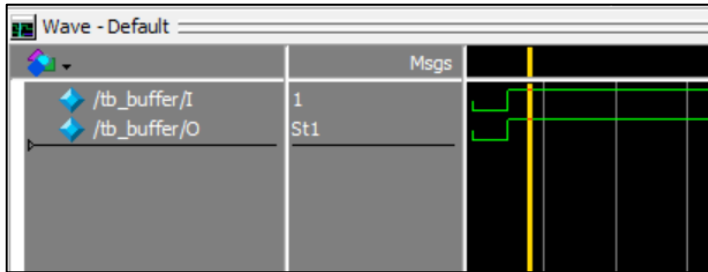
DUT:

```
buffer_test.v  tb_buffer.v  or_using_nand.v  tb_OR.v
1  module buffer(I, O);
2      input I;
3      output O;
4
5      buf b(O, I);
6
7  endmodule
8
```

Test Bench:

```
buffer_test.v  tb_buffer.v  or_using_nand.v  tb_OR.v
1  module tb_buffer();
2
3      reg I;
4      wire O;
5
6      buffer b(I, O);
7
8      initial begin
9          $monitor("I = %b O = %b", I, O);
10         I = 0;
11
12         #10 I = 1;
13
14     end
15
16 endmodule
```

Output:



```
sim x
Transcript
VSIM 4> run
# I = 0 O = 0
# I = 1 O = 1
VSIM 5>
```

Task 2:

Implement an inverter at the gate level.

Code:

DUT:

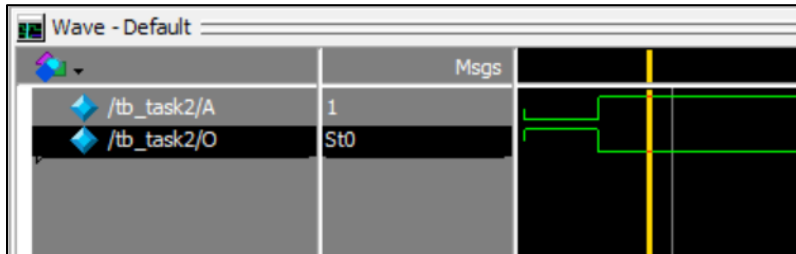
```
1 module task2(O, A);
2
3     input A;
4     output O;
5
6     not n1(O, A);
7
8 endmodule
```

```
1 module task2_nand(O, A);
2
3     input A;
4     output O;
5     nand n1(O, A);
6
7 endmodule
```

Test Bench:

```
1 module tb_task2();
2     reg A;
3     wire O;
4     task2 my_task2(O, A);
5
6     initial begin
7         $monitor("A = %b O = %b", A, O);
8
9         A = 0;
10        #10 A = 1;
11    end
12
13 endmodule
```

Output:



```
VSIM 13> run
# A = 0 O = 1
# A = 1 O = 0
```

Task 3:

Implement an OR gate using a NAND gates.

Code:

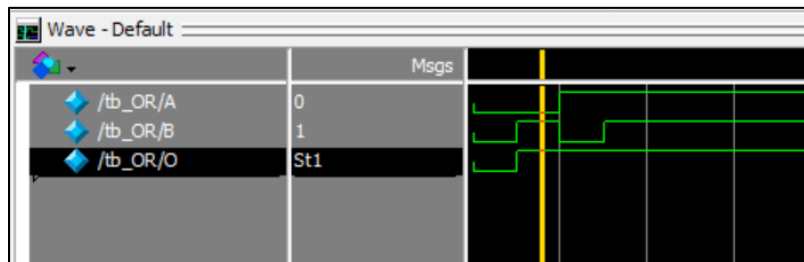
DUT:

```
buffer_test.v x tb_buffer.v x or_using_nand.v x tb_OR.v x
1  module or_using_nand(O, A, B);
2
3      input A, B;
4      output O;
5      wire nA, nB;
6
7      nand n3(O, nA, nB);
8      nand n1(nA, A);
9      nand n2(nB, B);
10
11  endmodule
```

Test Bench:

```
1  module tb_OR();
2
3      reg A, B;
4      wire O;
5      or_using_nand my_or_using_nand(O, A, B);
6
7      initial begin
8
9          $monitor("A = %b B = %b O = %b", A, B, O);
10
11         A = 0; B = 0;
12         #10 A = 0; B = 1;
13         #10 A = 1; B = 0;
14         #10 A = 1; B = 1;
15
16     end
17
18 endmodule
```

Output:



Task 4:

Implement the following equation where z is output and x_1, x_2, x_3, x_4 , and x_5 are inputs of the circuit.

$$z = (y_1 + y_2)'$$

$$y_1 = x_1.x_2 \quad y_2 = (x_3.x_4.x_5)$$

Code:

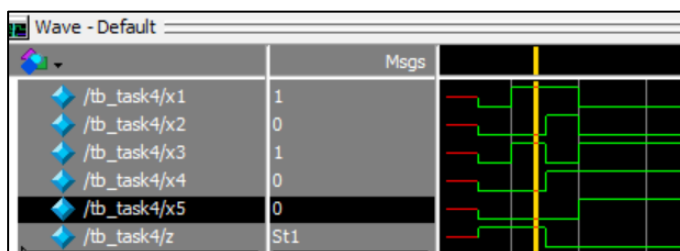
DUT:

```
1  module task4(z, x1, x2, x3, x4, x5);
2
3      input x1, x2, x2, x3, x4, x5;
4      output z;
5      wire y1, y2, zp;
6
7      and a1(y1, x1, x2);
8      and a2(y2, x3, x4, x5);
9
10     or o1(zp, y1, y2);
11     not n1(z, zp);
12
13 endmodule
```

Test Bench:

```
1  module tb_task4();
2      reg x1, x2, x3, x4, x5;
3      wire z;
4      task4 my_task4(z, x1, x2, x3, x4, x5);
5
6      initial begin
7          $monitor("x1 = %b, x2 = %b, x3 = %b, x4 = %b, x5 = %b, z = %b", x1, x2, x3, x4, x5, z);
8
9          #10 {x1, x2, x3, x4, x5} = 5'b00000;
10         #10 {x1, x2, x3, x4, x5} = 5'b10100;
11         #10 {x1, x2, x3, x4, x5} = 5'b11010;
12         #10 {x1, x2, x3, x4, x5} = 5'b00111;
13
14     end
15
16 endmodule
```

Output:



```
Transcript
VSIM 39> run
# x1 = x, x2 = x, x3 = x, x4 = x, x5 = x, z = x
# x1 = 0, x2 = 0, x3 = 0, x4 = 0, x5 = 0, z = 1
# x1 = 1, x2 = 0, x3 = 1, x4 = 0, x5 = 0, z = 1
# x1 = 1, x2 = 1, x3 = 0, x4 = 1, x5 = 0, z = 0
# x1 = 0, x2 = 0, x3 = 1, x4 = 1, x5 = 1, z = 0
```